

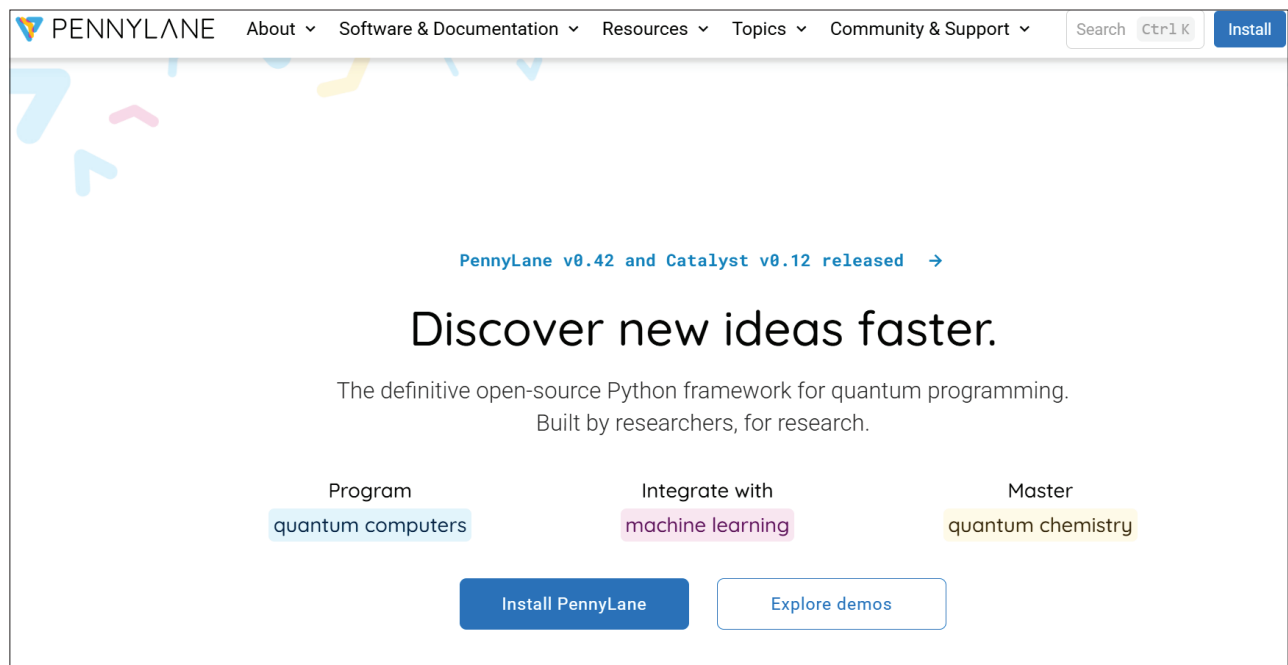


Figure 2: PennyLane for quantum programming

Table 2: Free and open source libraries for QML

Library	Programming language	URL
PennyLane	Python	pennylane.ai
Qiskit Machine Learning	Python	qiskit.org/ecosystem/machine-learning
TensorFlow Quantum	Python	tensorflow.org/quantum
TorchQuantum	Python	github.com/mit-han-lab/torchquantum
Cirq	Python	quantumai.google/cirq
MindQuantum	Python	github.com/mindspore-ai/mindquantum
Yao.jl	Julia	github.com/QuantumBFS/Yao.jl
QuTiP	Python	qutip.org
lambeq	Python	github.com/CQCL/lambeq
QuCumber	Python	github.com/PIQuIL/QuCumber
NetKet	Python	netket.org
OpenFermion	Python	github.com/quantumlib/OpenFermion
QGOpt	Python	github.com/LuchnikovI/QGOpt
Tequila	Python	github.com/tequilahub/tequila
Qibo	Python	qibo.science
D-Wave Ocean (dimod, dwave-networkx)	Python	docs.ocean.dwavesys.com
Strawberry Fields	Python	strawberryfields.ai
ProjectQ	Python	projectq.ch